13

Electromagnetic Induction

13.1

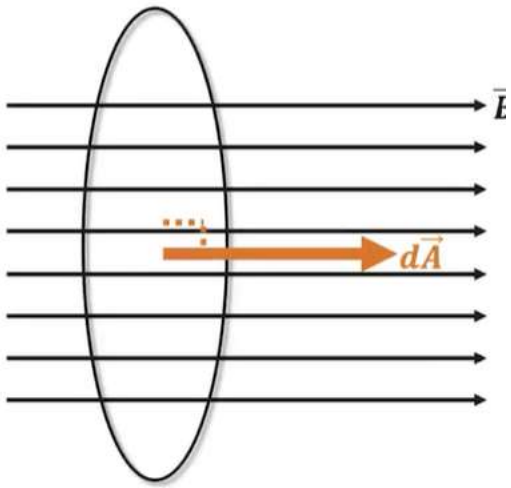
Magnetic Flux

13.1.1

Conceptual Magnetic Flux and Vector Fields

FLUX

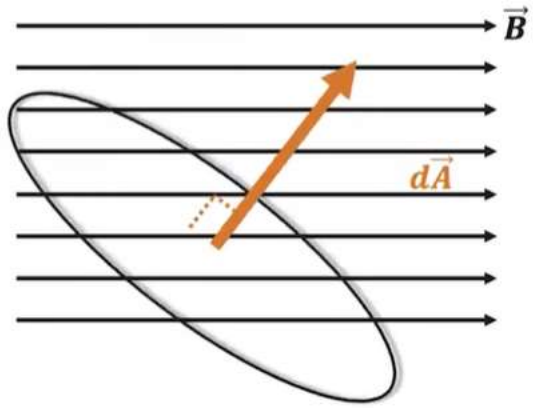
- Flux is an amount of stuff going through an area
- We've seen electric flux through a surface area: $\oint \vec{E} \cdot d\vec{A}$
- Now we have magnetic flux through a loop area: $\int \vec{B} \cdot d\vec{A}$
- The $\mathrm{d}\vec{A}$ vector points **perpendicular** to the area of the loop



FLUX

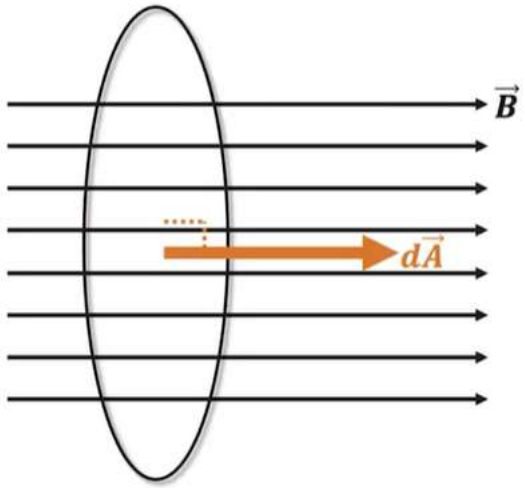
Remember dot product!

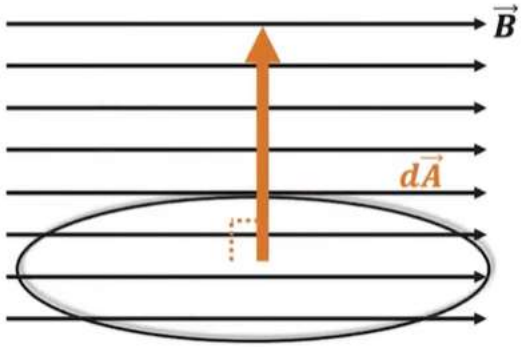
$$\vec{A} \cdot \vec{B} = |\vec{A}||\vec{B}| \cos\theta$$



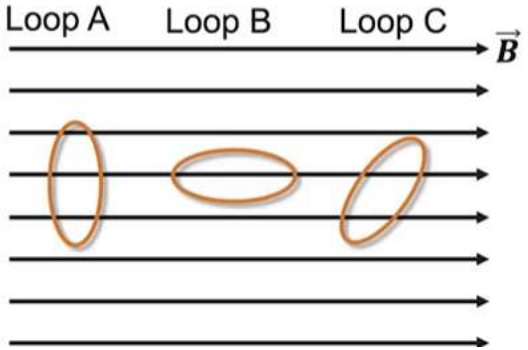
If $d\vec{A} \parallel \vec{B}$, then $\vec{B} \cdot d\vec{A} = |\vec{B}||d\vec{A}|$

If $d\vec{A} \perp \vec{B}$, then $\vec{B} \cdot d\vec{A} = 0$



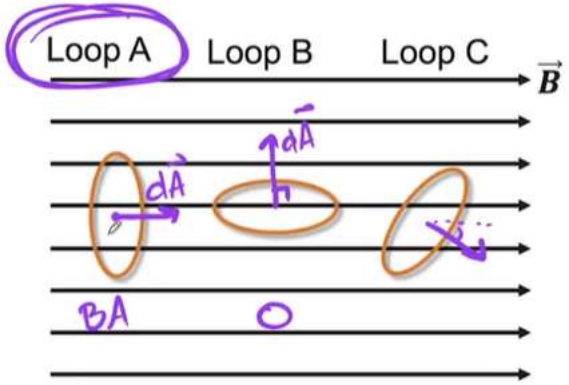


Which loop has greater magnetic flux?



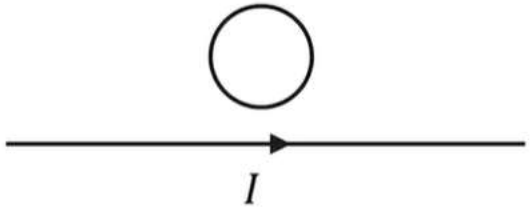
Which loop has greater magnetic flux?

$$\phi_B = \int \underbrace{\vec{B} \cdot d\vec{A}}_{B\,dA\cos\theta}$$



A small, circular loop of conducting wire is near a long, straight wire that carries a steady electric current I to the right, as shown below.

Which of the following actions will result in an increase in magnetic flux through the wire loop?



1. The wire loop is pulled vertically upward in the plane of the page (away from the wire)
2. The current in the straight wire is steadily increasing.
3. The wire loop is rotated about a diameter through a small angle so that it is no longer in the plane of the page.

A. 1 only

B. 2 only

C. 3 only

D. Either 1 or 2

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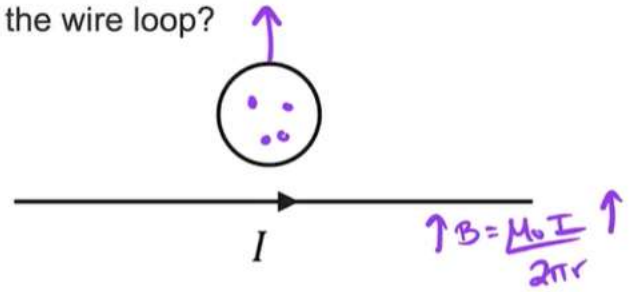
Which of the following actions will result in an increase in magnetic flux through the wire loop?



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A. 1 only

B. 2 only

C. 3 only

D. Either 1 or 2

Takeaways

Magnetic Flux depends on:

- Strength of Magnetic Field
- Size of loop area exposed to field
- Angle of loop in relation to field

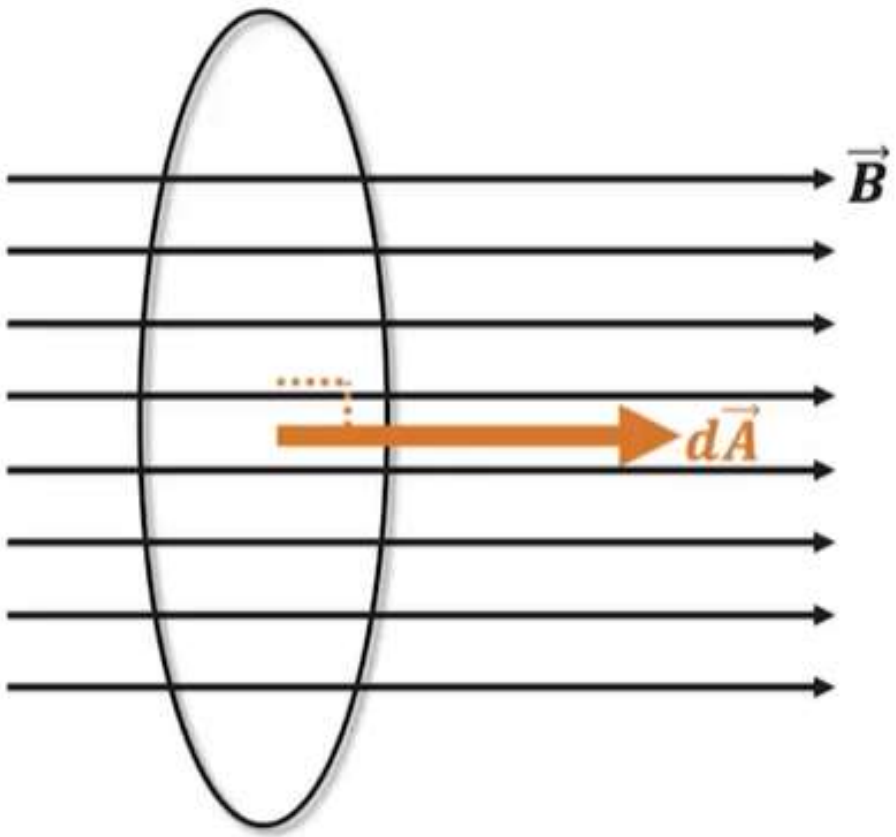
13.1.2

Calculating Common Situations of Magnetic Flux

Integrate or Not?

FLUX

- Flux is an amount of stuff going through an area
- We've seen electric flux through a surface area: $\oint \vec{E} \cdot d\vec{A}$
- Now we have magnetic flux through a loop area: $\int \vec{B} \cdot d\vec{A}$
- The $d\vec{A}$ vector points **perpendicular** to the area of the loop



FLUX

Remember dot product!

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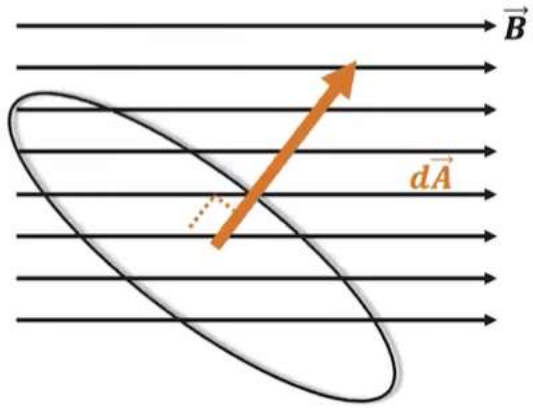

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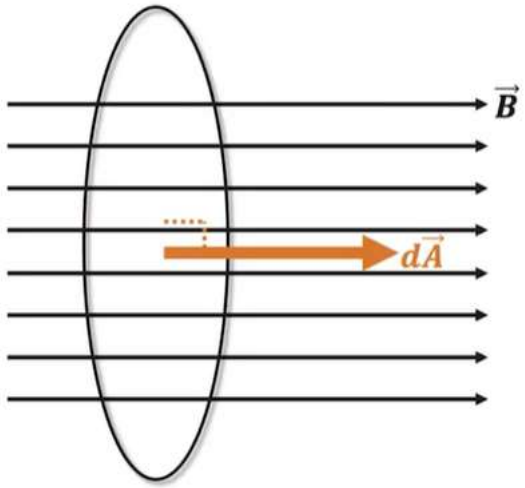
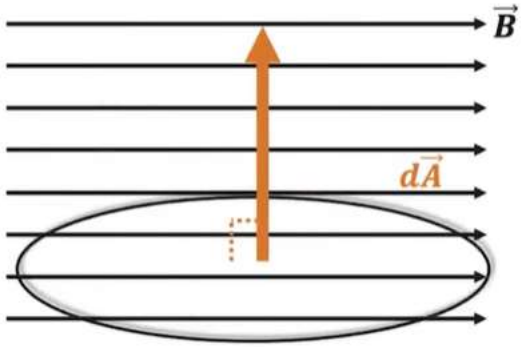




What is the flux through a hoop of radius R?

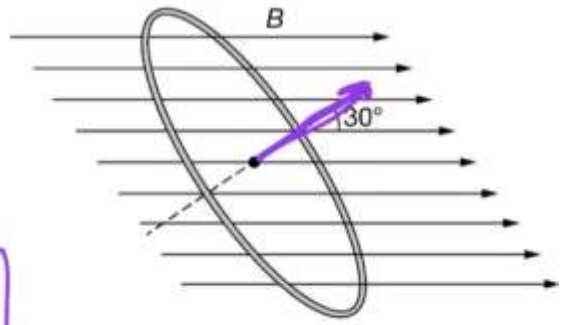
$$\Phi_B = \int \underbrace{\vec{B}\cdot d\vec{A}}_{B\,dA\cos\theta}$$

$$= \int B\,dA\cos\theta$$

$$= B\cos\theta \int dA$$

$$= B\cos\theta\, A$$

$$= B\cos 30\,\pi R^2 = \boxed{\frac{\sqrt{3}}{2} B\pi R^2}$$



What is the flux through a hoop of radius R?

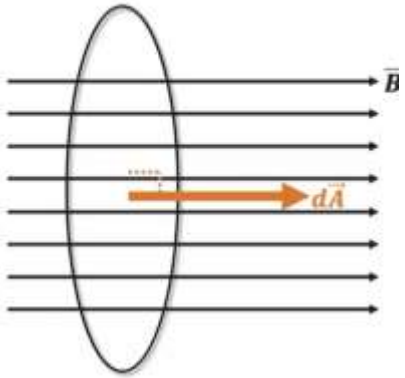
$$\Phi_B = \int \underbrace{\vec{B}\cdot d\vec{A}}_{\vec{B}\parallel d\vec{A}}$$

$$= \int B\,dA$$

$$= B\int dA$$

$$= BA$$

$$= \boxed{B\pi R^2}$$



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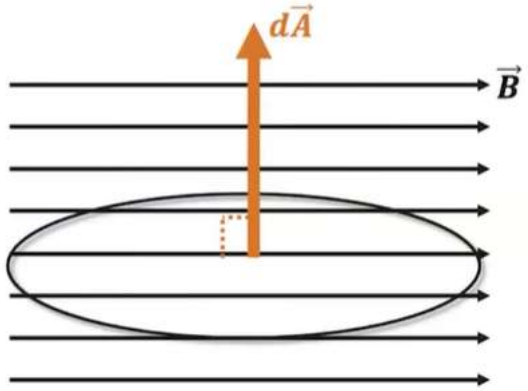
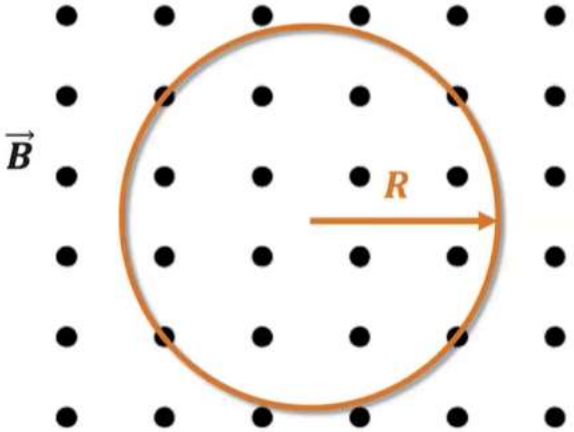


Magnetic flux with changing area

A circular current-carrying loop lies so that the plane of the loop is perpendicular to a constant magnetic field of strength $\vec{B}$.

Suppose that the radius R of the loop could be made to increase with time t so that $R = at$, where a is a constant.

What is the magnetic flux through the loop?

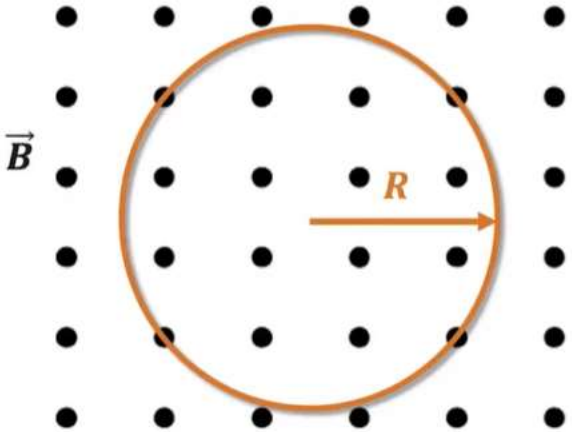


Source: Released AP Exam. Taken (and modified) from AP Classroom

Magnetic flux with changing area

$R = at$, where a is a constant.

What is the magnetic flux through the loop?



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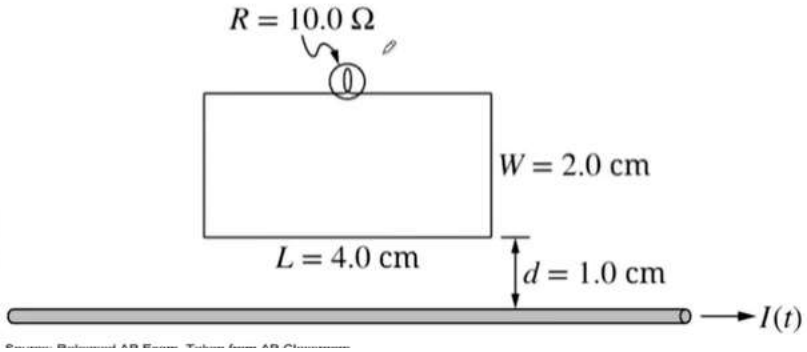
Loop and Wire FRQ

A lightbulb of **resistance $R = 10\ \Omega$** is connected to a rectangular loop of wire of negligible resistance near a very long current-carrying wire.

The rectangular loop has **length $L = 4.0\ \text{cm}$** and a **width of $w = 2.0\ \text{cm}$** and is positioned so one of the longer sides of the loop is a **distance $d = 1.0\ \text{cm}$** above and parallel to the long wire, as shown.

The current in the long wire is initially flowing to the right and is given by $I = C - Dt$, where $C = 10.0\ A$ and $D = 2.0\ A/s$.

At time $t = 5.0\ s$, the current in the long wire is instantaneously zero as the current changes direction.



Loop and Wire FRQ

Calculate the magnetic flux through the loop due to only the long wire at time $t = 3.0s$.

$I = C - Dt$

$B = \frac{\mu_0 I}{2\pi r}$

$\Phi_B = \int \underbrace{\vec{B} \cdot d\vec{A}}$ $\quad \vec{B} \parallel d\vec{A}$

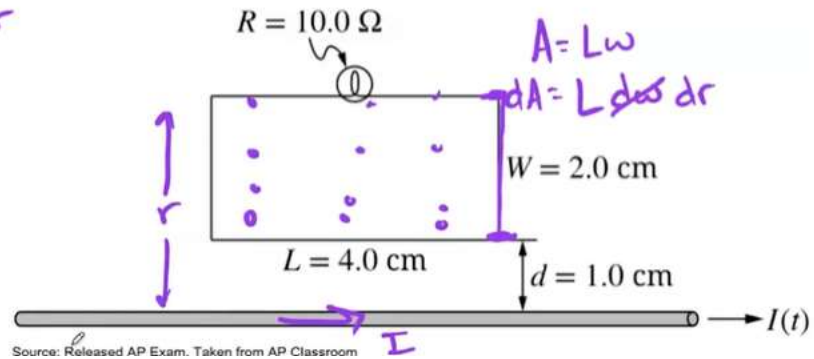
$= \int B\,dA = \int \frac{\mu_0 I}{2\pi r} L\,dr$

$= \frac{\mu_0 I L}{2\pi} \int_d^{d+w} \frac{1}{r}\,dr$

$= \frac{\mu_0 I L}{2\pi} \ln|r| \Big|_d^{d+w}$

$A = Lw$

$dA = L\,\cancel{dw}\,dr$



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Loop and Wire FRQ

Calculate the magnetic flux through the loop due to only the long wire at time $t = 3.0s$.

$I = C - Dt$

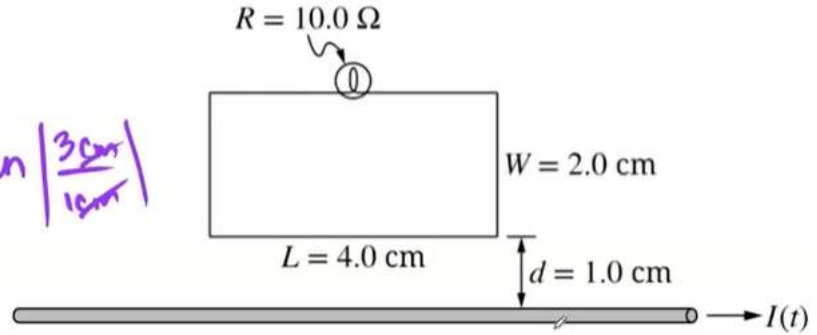
$10 - 2(3) = 4A$

$\Phi_B = \frac{\mu_0 I L}{2\pi} \ln|r| \Big|_d^{d+w}$

$\boxed{\Phi_B = \frac{\mu_0 I L}{2\pi} \ln\left|\frac{d+w}{d}\right|}$

$= \frac{(4\pi \times 10^{-7})(4A)(.04m)}{2\pi} \ln\left|\frac{3cm}{1cm}\right|$

$= \boxed{3.52 \times 10^{-8}\ T \cdot m^2}$



Source: Released AP Exam. Taken from AP Classroom

Takeaways

When to integrate for flux:

You only need to integrate if the magnetic field isn't constant over the entire area (it depends on r, or width, etc)

13.2

Electromagnetic Induction

13.2

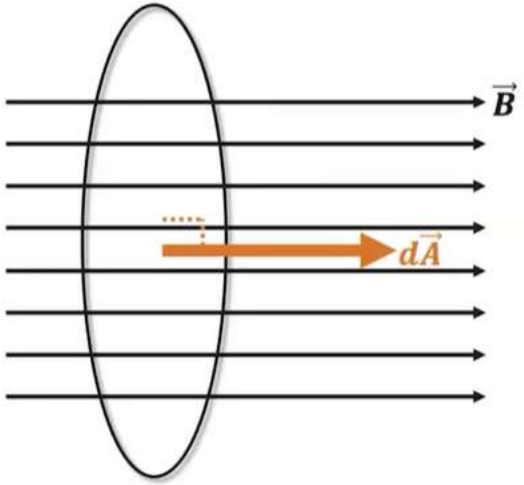
Electromagnetic Induction

13.2.1

Faraday’s Law Component Examples

MAGNETIC FLUX

- Now we have magnetic flux through a loop area: $\int \vec{B} \cdot d\vec{A}$
- The $d\vec{A}$ vector points **perpendicular** to the area of the loop
- Two ways to change Magnetic Flux: Change the field, or change the area exposed to the field (size or orientation/angle)



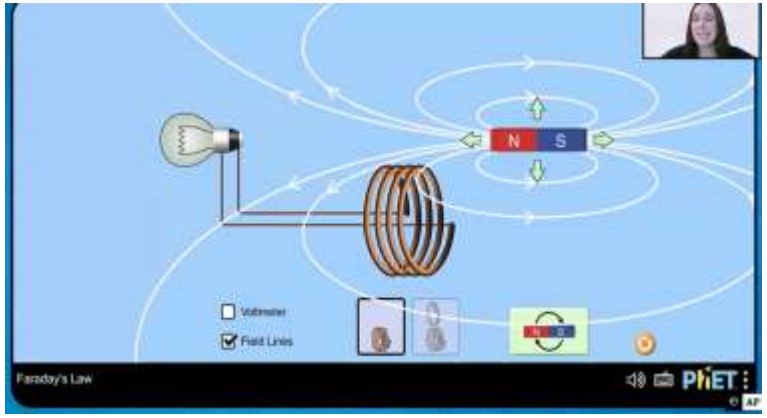
Ohm's Law:

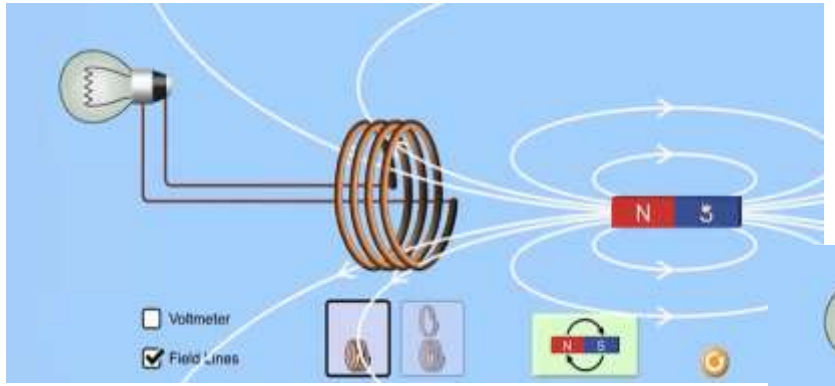
$$\mathbf{I} = \frac{\Delta \boldsymbol{V}}{\boldsymbol{R}}$$

Faraday's Law:

A changing magnetic flux induces an emf in the system

$$emf = -\frac{d\Phi_B}{dt}$$


N
S
Voltmeter
Field Lines
Faraday's Law
PhET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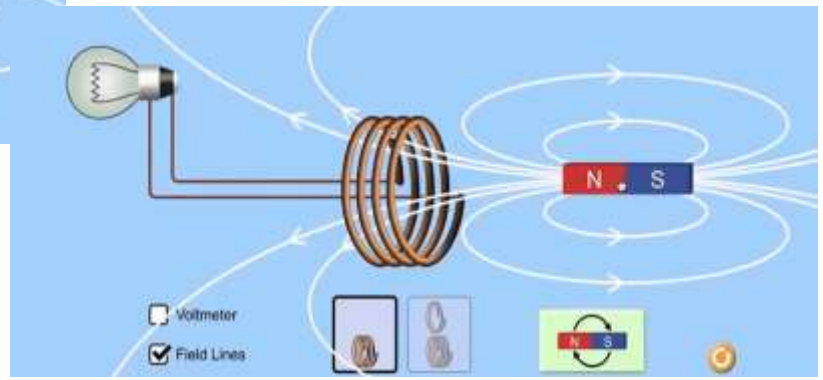
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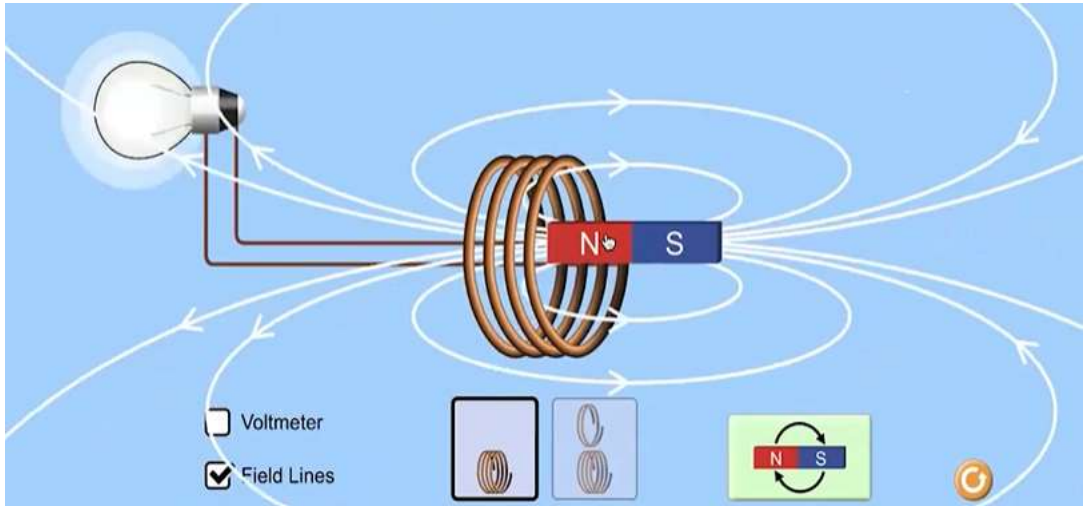
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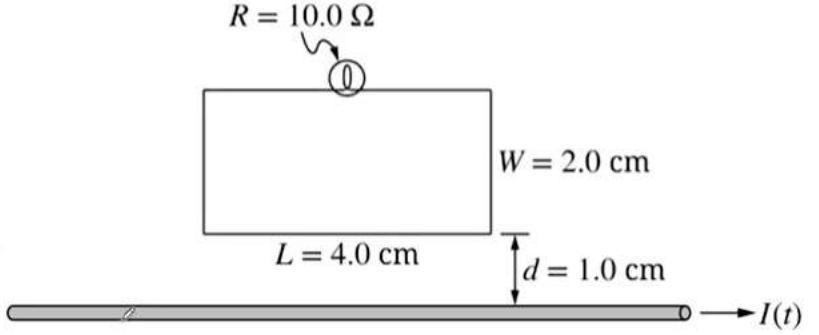


Source: Released AP Exam. Taken from AP Classroom

Loop and Wire FRQ

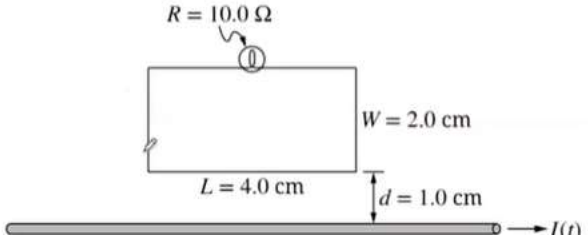
Calculate the magnitude of the induced emf generated around the loop at time $t = 3.0s$.

$$emf = -\frac{d\Phi_B}{dt} \qquad \Phi_B = \frac{\mu_0 I L}{2\pi} \ln\left(\frac{d+w}{d}\right) \qquad I = (C - Dt)$$

$$emf = -\left[\frac{\mu_0 L}{2\pi} \ln\left|\frac{d+w}{d}\right| (-D)\right] = \frac{\mu_0 L}{2\pi} \ln\left|\frac{d+w}{d}\right| (C - Dt)$$

$$\boxed{emf = \frac{\mu_0 L}{2\pi} \ln\left|\frac{d+w}{d}\right| D}$$

$$= \boxed{1.758 \times 10^{-8}\ V}$$



A circular current-carrying loop lies so that the plane of the loop is perpendicular to a constant magnetic field of strength B. Suppose that the radius R of the loop could be made to increase with time t so that $R = at$, where a is a constant.

What is the magnitude of the emf that would be generated around the loop as a function of t?

$\Phi_B = B\pi a^2 t^2$

$emf = -\frac{d\Phi_B}{dt} = -2B\pi a^2 t$

$\downarrow$

$\boxed{2B\pi a^2 t}$

A $2\pi B a^2 t$

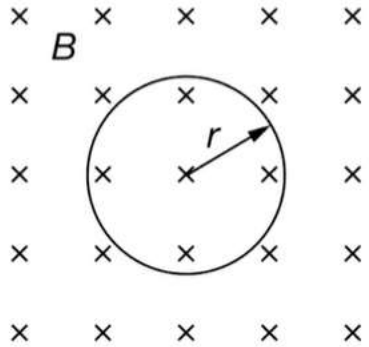
B $2\pi B a t$

C $2\pi B t$

D $\pi B a^2 t$

A single loop of wire is at rest in a magnetic field that is directed into the page, as shown in the figure below. The loop has a radius of r and a resistance of R. The magnetic field has a magnitude B that changes as a function of time t according to the equation $B = \alpha t + \beta t^2$, where α and β are both positive constants in units of T/s and T/s^2, respectively.

Which of the following expressions gives the current in the loop as a function of time t?



Source: Released AP Exam. Taken from AP Classroom

A $\dfrac{\beta t \pi r^2}{R}$

B $\dfrac{(\alpha + 2\beta t)\pi r^2}{R}$

C $\dfrac{(\alpha + \beta t) t \pi r^2}{R}$

D $\dfrac{(\alpha + 2\beta t) 2\pi r}{R}$

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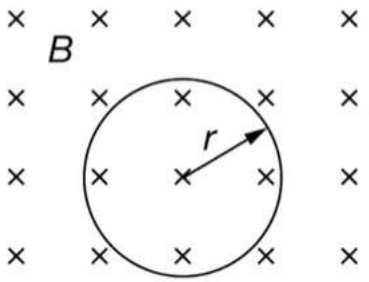
Which of the following expressions gives the current in the loop as a function of time t?

$$\text{emf} = -\frac{d\Phi_B}{dt}$$

$$\Phi_B = \left(\int \vec{B} \cdot d\vec{A}\right) = BA$$

$$= (\alpha t + \beta t^2)\pi r^2$$



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Which of the following expressions gives the current in the loop as a function of time t?

$\Phi_B = (\alpha t + \beta t^2)\pi r^2$

$emf = -\frac{d\Phi_B}{dt} = -(\alpha + 2\beta t)\pi r^2$

$I = \frac{\Delta V}{R} = \boxed{\frac{(\alpha + 2\beta t)\pi r^2}{R}}$

A $\frac{\beta t \pi r^2}{R}$

B $\frac{(\alpha + 2\beta t)\pi r^2}{R}$

C $\frac{(\alpha + \beta t)t\pi r^2}{R}$

D $\frac{(\alpha + 2\beta t)2\pi r}{R}$

Takeaways

A changing magnetic flux induces an emf in the system

Change the area (size or orientation/angle)

Change the field

13.2.2

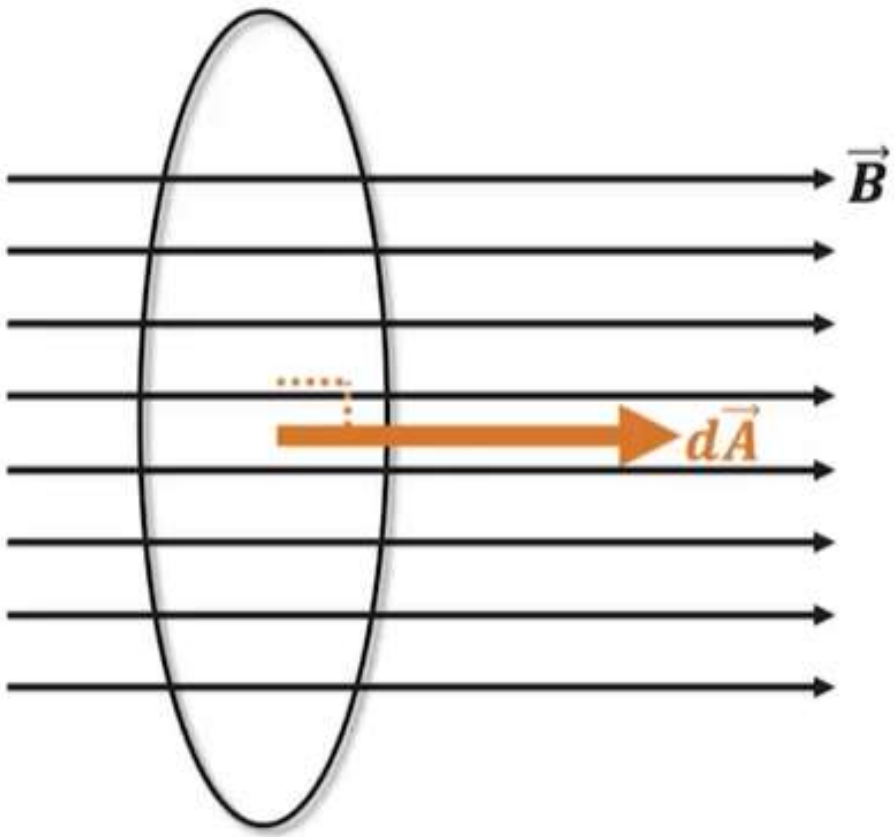
Lenz’s Law for Direction of Induced Current

-Mathematical Relationships

- “Arrows and Feathers”

FLUX

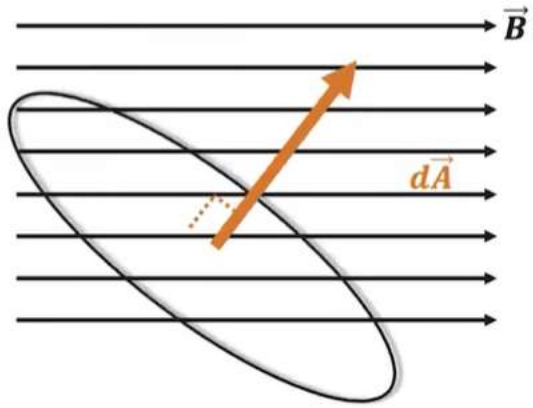
- Flux is an amount of stuff going through an area
- We've seen electric flux through a surface area: $\oint \vec{E} \cdot d\vec{A}$
- Now we have magnetic flux through a loop area: $\int \vec{B} \cdot d\vec{A}$
- The $d\vec{A}$ vector points **perpendicular** to the area of the loop



FLUX

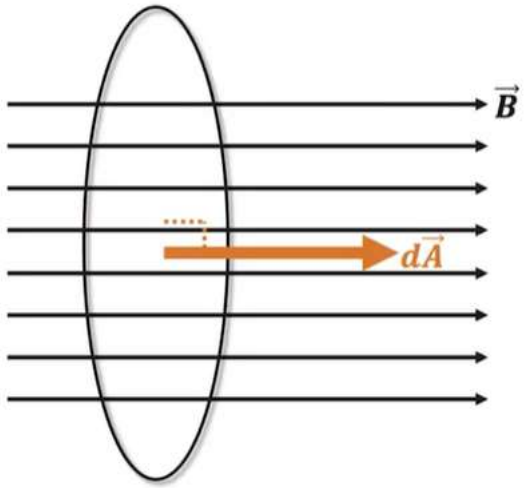
Remember dot product!

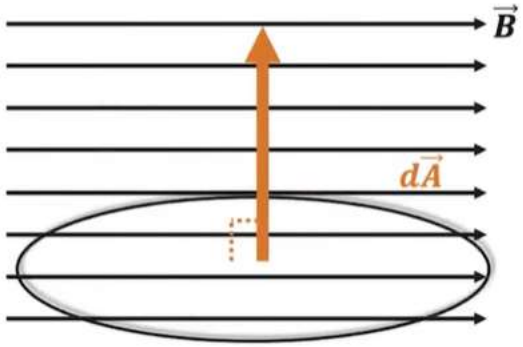
$$\vec{A} \cdot \vec{B} = |\vec{A}||\vec{B}| \cos\theta$$



If $d\vec{A} \parallel \vec{B}$, then $\vec{B} \cdot d\vec{A} = |\vec{B}||d\vec{A}|$

If $d\vec{A} \perp \vec{B}$, then $\vec{B} \cdot d\vec{A} = 0$





Faraday's Law:

A changing magnetic flux induces an emf in the system

$$emf = -\frac{d\Phi_B}{dt}$$

Lenz's Law:

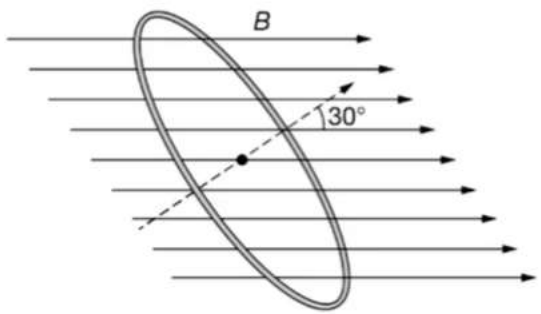
An induced emf generates a current that creates a magnetic field that opposes the change in flux

NATURE DOESN’T LIKE

CHANGE

A uniform magnetic field of magnitude $\boldsymbol{B}$ directed to the right exists in a certain region of space, as shown. A loop of wire is placed in the field such that a vector normal to the plane of the loop is at a **30° angle** relative to the field.

Which of the following is a correct statement about the direction of the current induced in the loop? emf = 0



A. The current starts clockwise and then becomes counterclockwise.

B. The current starts counterclockwise and then becomes clockwise.

C. The current is clockwise.

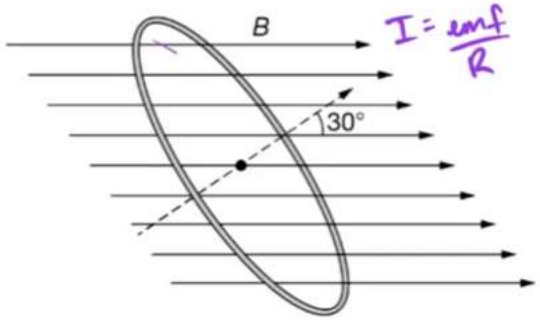
D. No current will be induced in the loop.

A uniform magnetic field of magnitude $\boldsymbol{B}$ directed to the right exists in a certain region of space, as shown. A loop of wire is placed in the field such that a vector normal to the plane of the loop is at a **30° angle** relative to the field.

Which of the following is a correct statement about the direction of the current induced in the loop?

$emf = -\frac{d\Phi_B}{dt}$

$I = \frac{emf}{R}$



A. The current starts clockwise and then becomes counterclockwise.

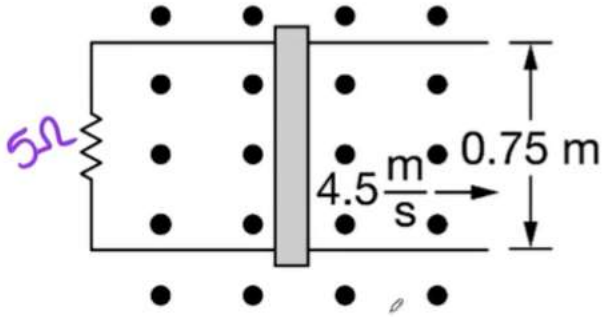
B. The current starts counterclockwise and then becomes clockwise.

C. The current is clockwise.

D. No current will be induced in the loop.

A conducting rod is sliding at a speed of $4.5\ \mathrm{m/s}$ along conducting rails that are $0.75\ \mathrm{m}$ apart. The rails are attached to a $5.0\ \Omega$ resistor, completing a loop. The loop is in a $250\ \mathrm{mT}$ magnetic field that is directed out of the page, as shown.

The current generated in the resistor is…



A) 0

B) $0.17\ A$ counterclockwise

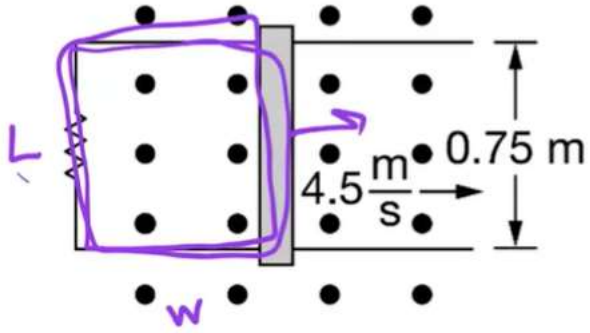
C) $0.17\ A$ clockwise

D) $0.84\ A$ clockwise

A conducting rod is sliding at a speed of $4.5\ \mathrm{m/s}$ along conducting rails that are $0.75\ \mathrm{m}$ apart. The rails are attached to a $5.0\ \Omega$ resistor, completing a loop. The loop is in a $250\ \mathrm{mT}$ magnetic field that is directed out of the page, as shown.

The current generated in the resistor is…

$A = LW$



$$\Phi_B = \int \vec{B}\, d\vec{A} = BA = BLW$$

$$\mathrm{Emf} = -\frac{d\Phi_B}{dt} = -\frac{d}{dt}(BLW)$$

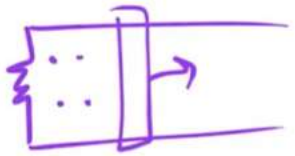
$$\mathrm{emf} = -BL\frac{dw}{dt} = -BLv$$

$$I = \frac{\Delta V}{R} = \frac{-BLv}{R}$$

$$I = \frac{(250\times10^{-3}\ \mathrm{T})(.75\ \mathrm{m})(4.5\ \mathrm{m/s})}{5\ \Omega}$$

A conducting rod is sliding at a speed of $4.5\ \mathrm{m/s}$ along conducting rails that are $0.75\ \mathrm{m}$ apart. The rails are attached to a $5.0\ \Omega$ resistor, completing a loop. The loop is in a $250\ \mathrm{mT}$ magnetic field that is directed out of the page, as shown.

The current generated in the resistor is…



$\mathrm{emf} = -\frac{d\Phi_B}{dt}$

induce

A. ~~0~~

B. $0.17\ A$ counterclockwise

C. $0.17\ A$ clockwise

D. ~~$0.84\ A$ clockwise~~

Faraday's Law:

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$$emf = -\frac{d\Phi_B}{dt}$$

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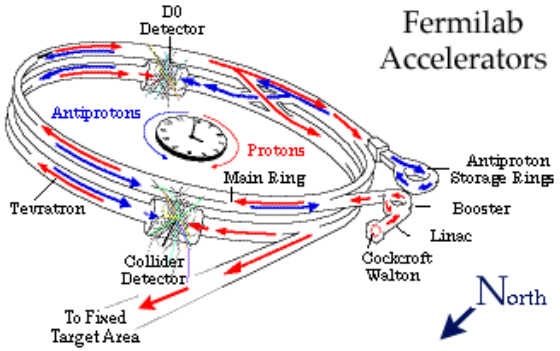
An induced emf generates a current that creates a magnetic field that opposes the change in flux

NATURE DOESN’T LIKE

CHANGE

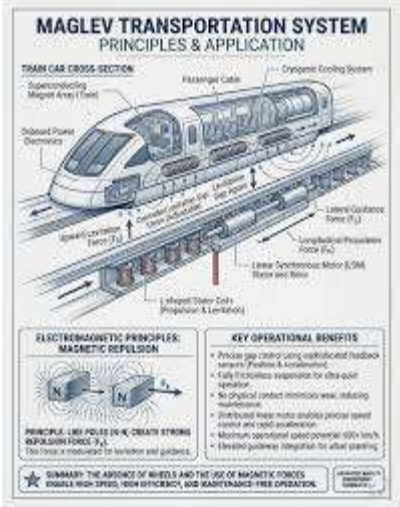
13.2.3

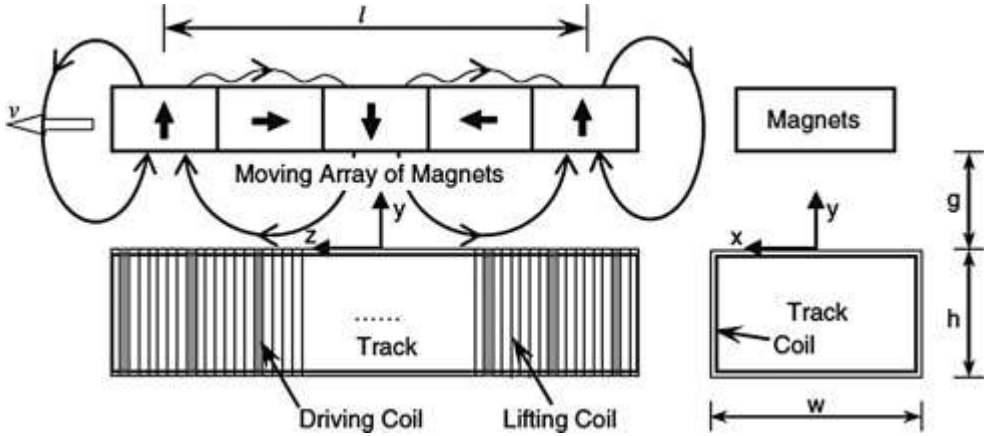
Maxwell’s Equations for Electric and Magnetic Fields

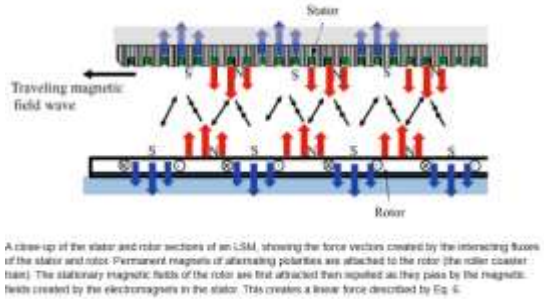


13.2.3

Maxwell's Equations for Electric and Magnetic Fields







Maxwell's Equations:

Maxwell's Equations are the set of equations that fully describe electromagnetism

Maxwell's First Equation:

Electric Charges
-> Electric Fields

Gauss's Law

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\varepsilon_0}$$

Our Version

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\varepsilon_0}$$

Future Usage

Gradient Vector

"Del (Delta) Operator"

The changing field will either
Approximate/ Behave like
-A Collapsing field <u>towards</u> a negative charge

OR
-An emerging field <u>from</u> a positive charge

Maxwell's Second Equation:

Gauss's Law for Magnetism

Magnetic Field Lines always form a closed loop,

So, the magnetic flux through a closed 3D surface is always zero

$$\oint \vec{B} \cdot d\vec{A} = 0$$

Our Version

$$\vec{\nabla} \cdot \vec{B} = 0$$

Future Usage

Gradient Vector

"Del (Delta) Operator"

The changing field will either
Approximate/ Behave like
-Field Lines emerging from the north pole and wrapping around to the south pole

-NO Magnetic Monopoles, Only Dipoles (N or S)

Maxwell's Third Equation:

Faraday's Law

$$emf = \oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial B}{\partial t}$$

Maxwell's Fourth Equation:

Ampere's Law

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I + \mu_0 \varepsilon_0 \frac{d\Phi_E}{dt}$$

$$\vec{\nabla} \times \vec{B} = \mu_0 (\vec{J} + \varepsilon_0 \frac{\partial E}{\partial t})$$

Maxwell's Fourth Equation:

Ampere's Law

$$\boxed{\oint \vec{B} \cdot d\vec{l} = \mu_0 I} + \mu_0 \varepsilon_0 \frac{d\Phi_E}{dt}$$

Maxwell's Addition

$$\vec{\nabla} \times \vec{B} = \mu_0 (\vec{J} + \varepsilon_0 \frac{\partial E}{\partial t})$$

Magnetic Fields can be created by either:

- A current

- Changing electric field

Takeaways

$$\oint \vec{E} \cdot d\vec{A} = \frac{\cancelto{0}{Q_{enc}}}{\varepsilon_0}$$

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$$

$$\oint \vec{B} \cdot d\vec{A} = 0$$

$$\oint \vec{B} \cdot d\vec{l} = \cancelto{0}{\mu_0 I} + \mu_0 \varepsilon_0 \frac{d\Phi_E}{dt}$$

*In Empty Space with ≈ 0 Fields

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RADIATIVE TRANSFER IN A CLUMPY UNIVERSE. IV. NEW SYNTHESIS MODELS OF THE COSMIC UV/X-RAY BACKGROUND

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ABSTRACT

We present improved synthesis models of the evolving spectrum of the UV/X-ray diffuse background, updating and extending our previous results. Five new main components are added to our radiative transfer code CUBA: (1) the sawtooth modulation of the background intensity from resonant line absorption in the Lyman series of cosmic hydrogen and helium; (2) the X-ray emission from the obscured and unobscured quasars that gives origin to the X-ray background; (3) a piecewise parameterization of the distribution in redshift and column density of intergalactic absorbers that fits recent measurements of the mean free path of 1 ryd photons; (4) an accurate treatment of the photoionization structure of absorbers, which enters in the calculation of the helium continuum opacity and recombination emissivity; and (5) the UV emission from star-forming galaxies at all redshifts. We provide tables of the predicted H I and He II photoionization and photoheating rates for use, e.g., in cosmological hydrodynamics simulations of the Lyα forest and a new metallicity-dependent calibration to the UV luminosity density-star formation rate density relation. A "minimal cosmic reionization model" is also presented in which the galaxy UV emissivity traces recent determinations of the cosmic history of star formation, the luminosity-weighted escape fraction of hydrogen-ionizing radiation increases rapidly with look-back time, the clumping factor of the high-redshift intergalactic medium evolves following the results of hydrodynamic simulations, and Population III stars and miniquasars make a negligible contribution to the metagalactic flux. The model provides a good fit to the hydrogen-ionization rates inferred from flux decrement and proximity effect measurements, predicts that cosmological H II (He III) regions overlap at redshift 6.7 (2.8), and yields an optical depth to Thomson scattering, $\tau_{es} = 0.084$ that is in agreement with *Wilkinson Microwave Anisotropy Probe* results. Our new background intensities and spectra are sensitive to a number of poorly determined input parameters and suffer from various degeneracies. Their predictive power should be constantly tested against new observations. We are therefore making our redshift-dependent UV/X emissivities and CUBA outputs freely available for public use at http://www.ucolick.org/~pmadau/CUBA.

Key words: cosmology: theory – diffuse radiation – intergalactic medium – galaxies: evolution – quasars: general

Online-only material: color figures

1. INTRODUCTION

The reionization of the all-pervading intergalactic medium (IGM) is a landmark event in the history of cosmological structure formation. Studies of Gunn–Peterson absorption in the spectra of distant quasars show that hydrogen is highly photoionized out to redshift $z \gtrsim 6$ (e.g., Fan et al. 2006a; Songaila 2004), while polarization data from the *Wilkinson Microwave Anisotropy Probe (WMAP)* constrain the redshift of a sudden reionization event to be significantly higher, $z = 10.5 \pm 1.2$ (Jarosik et al. 2011). It is generally thought that the IGM is kept ionized by the integrated UV emission from active nuclei and star-forming galaxies, but the relative contributions of these sources as a function of epoch are poorly known. Because of the high ionization threshold (54.4 eV) and 2001; Wyithe & Loeb 2003; Meiksin 2005; Trac & Cen 2007; Faucher-Giguère et al. 2008a; Gilmore et al. 2009; Robertson et al. 2010). This idea may be supported by the detection of escaping ionizing radiation from individual Lyman break galaxies at $z \sim 3$ (e.g., Shapley et al. 2006).

Despite much recent progress, a coherent description of the thermal state and ionization degree of the IGM remains elusive. The intensity and spectrum of the cosmic ultraviolet background remain one of the most uncertain yet critically important astrophysical input parameters for cosmological simulations of the IGM and for interpreting quasar absorption-line data and derive information on the distribution of primordial baryons (traced by H I, He I, He II transitions) and of the nucleosynthetic products of star formation (C III, C IV, Si III, Si IV, O VI, etc.). This is the fourth paper in a series aimed at a detailed study

IV. Cloudy Models

A. UV Background Only

We begin by establishing the HM12 ionization in the cloudy models. Its spectrum is shown below. We can then determine an appropriate cloud density such that the CGM is accurately represented in the models. As previously mentioned, this will be accomplished by varying the hydrogen density (n_H) of the cloudy models.

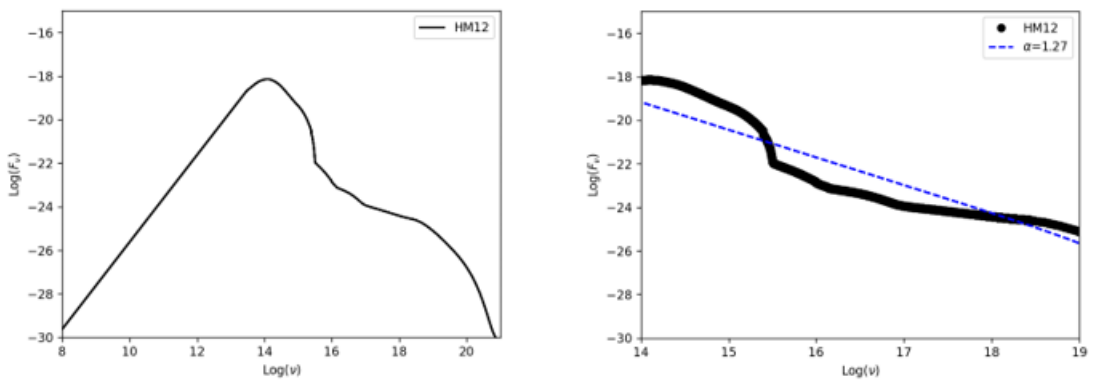

Figure 3: (a) Haardt & Madau (2012) spectrum. (b) Power law fit to UV region of HM12 spectrum. Note that our χ^2fit indicated $\alpha_{UVB} = 1.27$. Note that ν is in Hertz and F_ν is in ergs·cm^{-2}s^{-1}

Non- Zero Terms

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \varepsilon_0 \frac{d\Phi_E}{dt}$$

Magnetic Field
Opposes Direction!

"Response" E Field
OPPOSES the original field

Changing Electric Fields
Create a Magnetic Field

AND

Changing Magnetic Fields
Create Electric Fields

"EMF Waves" AKA LIGHT!

Speed of Light

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} \approx 3 \times 10^8 \, m/s$$

13

Electromagnetic Induction

13.3

Induced Currents and Magnetic Forces

13.3.1

The force exerted an external magnetic field on a conducting loop with an induced current